

Beyond Factual Queries: A Novel Predictive Retrieval-Augmented Generation

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Abstract

Retrieval-augmented generation (RAG) has proven effective at mitigating limitations of large language models (LLMs), including outdated knowledge, semantic gaps, and hallucinations. However, existing RAG techniques are primarily optimised for factual question answering—where answers can be directly retrieved—rather than for predictive tasks that require inferring unknown outcomes, such as user preferences in recommendations. To address this gap, we propose **PRAG**, a RAG framework tailored for predictive settings. PRAG computes the *semantic similarity* between a user’s historical interactions and target information, and adaptively integrates this similarity as prompt weights, thereby enhancing the LLM’s ability to model personalised, semantics-informed preferences. PRAG integrates seamlessly into LLM-based applications; instantiated in recommender systems, it explicitly captures user-specific preference signals via semantic similarity modelling. Experiments using two LLM backbones and four real-world datasets show that PRAG significantly improves predictive recommendation performance. Our source code is available at <https://github.com/djf-web/PRAG>.

CCS Concepts

• Computing methodologies → Lexical semantics; • Information systems → Recommender systems.

Keywords

Large Language Models, Semantics, Retrieval-Augmented Generation, Hallucinations

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1 Introduction

The rapid development of large language models (LLMs) has significantly propelled the proliferation of intelligent applications, and their exceptional language generation capabilities have been widely recognised [1, 59, 61]. LLMs have demonstrated remarkable success across a wide range of domains, including open-domain dialogue systems [53, 60], code generation [6, 54], and multimodal reasoning [2, 47]. Moreover, their ability to capture rich semantic representations and integrate knowledge has enabled breakthroughs in semantic understanding, knowledge-based question answering, and reasoning tasks [21, 63, 69]. However, LLMs still face two major challenges: first, their performance is limited in knowledge domains that are under-represented in the training data [42, 57]; second, they are prone to factual errors or “hallucinations” [29, 31, 52].

Retrieval-Augmented Generation (RAG) has been developed to enhance large language models (LLMs) by incorporating external retrieval systems. This technique dynamically acquires and integrates up-to-date, relevant knowledge, thereby effectively mitigating the “hallucination” problem caused by the static knowledge of models [39]. Recent research has made significant progress in RAG, focusing primarily on three directions [20]: (1) Retrieval Optimisation [18, 40, 51, 62, 66, 68, 72]: for example, RRR [51] leverages LLMs to rewrite queries for better semantic alignment; BGE [66] fine-tunes embedding models with domain knowledge or LLM feedback; and EventRAG [68] incorporates event knowledge graphs into retrieval. (2) Generation Optimisation [10, 33, 44, 46]: for example, Self-Mem [10] uses retrieved content for supervised training; SURGE [33] enhances generation consistency through graph-text contrastive learning; KAG [44] introduces a knowledge-augmented generation framework to improve the integration and utilisation of retrieved knowledge. (3) Augmentation process [3, 5, 9, 22, 30, 70]: for example, REALM [22] incorporates retrieval into the pre-training stage; UPRISE [9] optimises the retriever or generator during fine-tuning; Self-RAG [3] leverages reflection tokens to dynamically retrieve passages at inference time.

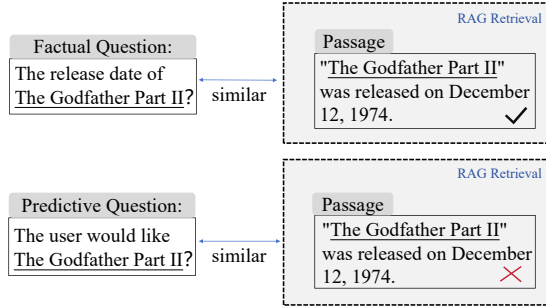


Figure 1: RAG can accurately match answers in factual question-answering scenarios, but fails to precisely match answers in predictive question-answering contexts.

Existing RAG methods, though highly effective in factual reasoning, are inherently unsuitable for predictive tasks—for instance, estimating whether a user will like an unseen item—since the “ground truth” does not exist beforehand and cannot be retrieved. As illustrated in Figure 1, in factual question answering (QA), for instance, the question “The release date of The Godfather Part II?”—RAG can accurately retrieve relevant external knowledge such as “The Godfather Part II was released on December 12, 1974.” However, in predictive QA—for example, the question “Would the user like The Godfather Part II?”, since user preferences are inherently unknown and cannot be directly retrieved from external knowledge, conventional RAG struggles to be effective.

Many predictive tasks in LLMs still struggle to effectively leverage external retrieval information to enhance model performance due to the limitations of traditional RAG techniques. For example, in recommendation scenarios, user preferences are typically unknown, making it difficult to directly apply existing RAG methods to improve recommendation accuracy. In recent years, LLM-based recommender systems have achieved remarkable progress; however, most approaches still primarily rely on user–item ID information [4, 11, 16, 41, 48, 55] or collaborative signals [13, 14, 19, 45, 71, 73, 75] for modelling. For instance, TALLRec [4] constructs fine-tuning instructions based on users’ historical interaction IDs to capture interaction semantics, while CoLLM [73] incorporates collaborative information during fine-tuning to improve performance under cold-start settings. However, these methods fail to explicitly model the semantic relevance between users’ historical behaviours and target items, thereby overlooking users’ personalised preference differences across items.

To address this challenge, we propose PRAG (Predictive Retrieval-Augmented Generation), a framework that adapts RAG for predictive scenarios. PRAG computes the semantic similarity between a user’s historical interactions and a target item, and adaptively incorporates these similarity values as *dynamic weights* into the prompt. This design explicitly encodes the semantic relationship between past and target items, enabling the LLM to model user-specific, semantics-informed preferences. PRAG integrates seamlessly into LLM-based applications, such as recommender systems, where it dynamically adjusts the influence of history–target similarity to enhance personalised modelling. Experimental results on two LLM

backbones (LLaMA-7B and Vicuna-7B) and four real-world datasets demonstrate that PRAG consistently improves predictive recommendation performance.

In summary, our main contributions are listed as follows:

- We identify the inherent limitation of traditional RAG methods in predictive scenarios, where answers cannot be retrieved because outcomes are unknown a priori (e.g., user preferences in recommendation tasks).
- We propose PRAG, a RAG framework tailored for predictive tasks, which employs a semantic similarity–driven dynamic weighting mechanism to explicitly encode user–target associations in prompts.
- We integrate PRAG into existing LLM-based recommender systems, resolving the mismatch between ID-based interest modelling and the semantic reasoning capabilities of LLMs, thus enabling semantics-aware preference modelling.
- Extensive experiments on two LLM backbones (LLaMA-7B and Vicuna-7B) and four real-world datasets demonstrate that PRAG consistently enhances predictive recommendation performance across diverse settings.

2 Related Work

In this section, we review related research from two main perspectives: (1) Retrieval-Augmented Generation (RAG), which provides the foundational paradigm our work builds upon, and (2) LLM-based recommender systems, where our proposed PRAG framework is applied and evaluated.

2.1 Retrieval Augmented Generation

RAG enhances task performance by integrating external knowledge into LLMs, thereby incorporating domain-specific information to improve accuracy and reliability [22, 35, 39, 49]. In recent years, extensive research on RAG has primarily focused on three optimisation dimensions: retrieval, generation, and augmentation [20]. (1) Retrieval Optimisation: Methods such as RRR [51], HyDE [18], and Query2Doc [62] leverage LLMs to rewrite queries for better alignment in semantic space; BGE [66] and LLM-Embedder [72] fine-tune embedding models using domain knowledge or LLM feedback. Additionally, retrieval has been extended to structured data such as images, code, and knowledge graphs (e.g., BLIP-2 [40], KnowledgeGPT [64]), along with the introduction of event knowledge graphs in EventRAG [68]. (2) Generation Optimisation: Approaches like Self-Mem [10] and RA-DIT [46] employ retrieved content for supervised training; SURGE [33] enhances generation consistency through graph-text contrastive learning; KAG [44] proposes a knowledge-augmented generation framework to improve the model’s ability to integrate and utilise retrieved knowledge. (3) Augmentation Process: REALM [22], RETRO [5], and Atlas [30] incorporate retrieval mechanisms during pre-training; REPLUG [56] and UPRISE [9] optimise retrievers or generators during fine-tuning; FLARE [32] and Self-RAG [3] perform dynamic retrieval during inference; DRAG [70] further optimises computational resource allocation under long-context conditions.

Despite the remarkable progress of RAG, these methods primarily target *factual* question answering—retrieving and verifying existing knowledge—rather than *predictive* tasks that involve

inferring unknown outcomes. To address this gap, our proposed PRAG framework extends RAG to predictive settings by introducing semantic similarity-driven prompt weighting, enabling LLMs to reason about latent, user-specific information. PRAG is seamlessly integrable with LLM-based applications, such as recommender systems, to improve predictive reasoning and personalisation.

2.2 LLM-based Recommender Systems

LLMs have recently emerged as a promising backbone for recommender systems, owing to their ability to capture semantic and contextual relationships across users and items [4, 7, 8, 73]. Existing research in this domain can be broadly categorised into two lines: *ID-based approaches* and *collaborative-aware approaches*.

(1) **ID-based LLM recommenders.** Several studies rely primarily on user-item identifiers to model user preferences. GPT4RS [11] explored various prompt designs that incorporate users' historical interaction IDs to improve recommendation performance. LLM-Rec [55] examined LLM-based recommendations in cold-start scenarios using ID information. ChatGPTGoodRec [48] systematically evaluated ChatGPT's recommendation ability across multiple domains. In addition, fine-tuning strategies have been adopted to further align LLMs with recommendation objectives. For example, TALLRec [4] fine-tuned LLMs with task-specific instruction data, E4SRec [41] combined LLMs with traditional recommender models for higher efficiency, and LLMCRS [16] applied reinforcement learning to enhance performance in conversational recommendation. However, relying solely on ID information limits the model's ability to capture collaborative signals and fine-grained semantic dependencies between users and items.

(2) **Collaborative and semantic-enhanced recommenders.** To better capture user-item relationships, recent studies have integrated collaborative or semantic information into LLM-based frameworks. Chat-Rec [19] incorporated user profiles and historical behaviours into prompts for interactive recommendation generation. LLaRA [45] developed a hybrid item representation that combines traditional recommender embeddings with LLM semantics. FaiRLLM [71] fine-tuned LLMs to produce high-quality textual representations for improved ranking. LC-Rec [75] jointly modelled linguistic and collaborative semantics during fine-tuning to enhance recommendation accuracy. CoLLM [73] further improved performance in warm-start scenarios by injecting collaborative information during fine-tuning.

While these approaches effectively capture ID-based and collaborative information, they often neglect personalised semantic signals that differ across users and items. Such omission restricts the LLM's ability to model nuanced, user-specific preferences and leads to suboptimal personalisation. Our proposed PRAG framework directly addresses this limitation by introducing a semantic similarity-weighted retrieval mechanism, enabling LLMs to adaptively model personalised and predictive user-item relationships.

As shown in Figure 2, whether a user will like *The Godfather Part II* may be strongly influenced by their history of watching *The Godfather*, while the impact of having watched *Titanic* is relatively minor. Existing LLM-based recommender systems typically feed a user's entire historical interaction sequence uniformly into the

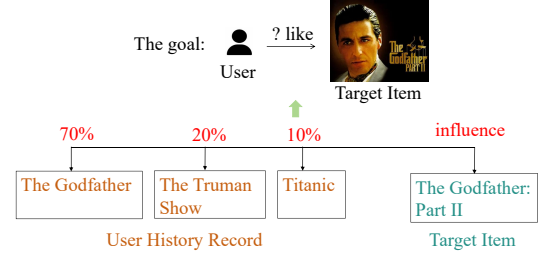


Figure 2: Differential influence of items in a user's historical records on their potential preference for a target item. For example, *The Godfather* in the user's history exerts substantially greater influence on their likelihood of liking *The Godfather Part II* compared to other items.

prompt of an LLMs. For example, if a user has watched *The Godfather*, *The Truman Show*, and *Titanic*, the system may prompt the model to determine whether the user will like *The Godfather Part II* – implying that all historical interactions are assigned equal weight. In real-world scenarios, however, a user's preference for certain items is often influenced more critically by a subset of their historical interactions. In contrast, by integrating PRAG technology into existing LLM-based recommender systems, we can more effectively model users' personalised interests information, thereby improving the recommendation accuracy of LLM-based recommender systems.

3 Preliminaries

3.1 Notations

In this work, we present the essential notations, which are outlined in Table 1.

Table 1: Definitions and Notations

Symbol	Definition
X	Instruction Input.
Y	Instruction Output.
Y_t	The t -th token of Y .
$Y_{<t}$	Tokens preceding Y_t .
$\mathcal{M}$	Large language model.
$\mathcal{Z}$	Training dataset.
Φ	Original parameters of the large model M .
Θ	LoRA parameters.

Let X denote the instruction inputs and Y is defined as instruction output. Y_t is the t -th token of Y , and $Y_{<t}$ represent the tokens preceding Y_t . Let Θ denote the LoRA (Low-Rank Adaptation) parameters, and Φ represent the original parameters of LLM.

3.2 Problem Statement

Given a user input X and an external knowledge base $\mathcal{D} = \{d_i\}_{i=1}^{|\mathcal{D}|}$, traditional RAG techniques aim to utilise information retrieved from the external knowledge base to generate an answer $Y = [S_1, S_2, \dots, S_m] = [W_1, W_2, \dots, W_n]$ consisting of m sentences or n tokens. In retrieval-augmented language models, the language model is typically paired with a retriever that can retrieve a list of external knowledge $\mathcal{D}_q = \text{ret}(q)$ for a query q . The language model then generates the answer based on the user input X and the retrieved external knowledge $\mathcal{D}_q$.

However, in predictive tasks, e.g., estimating whether a user will like an item—the ground-truth label is unknown at inference time, so conventional RAG cannot retrieve a definitive answer. To address this limitation, we introduce PRAG, a retrieval-augmented prompting strategy tailored for prediction. PRAG first computes the semantic similarity between a target instance and relevant historical records, and then injects these similarity signals into the prompt. By conditioning the LLM on retrieved evidence and its quantified relevance, PRAG improves the model's performance on predictive tasks where exact answers are not directly retrievable.

3.3 Instruction Tuning

Although LLMs (such as GPT, LLaMA, etc.) demonstrate remarkable capabilities in language understanding and generation, they are typically more focused on text processing rather than executing specific tasks like recommendations without specialised training. To enhance their performance in particular tasks, instruction tuning is often required. Instruction tuning refers to the retraining of large language models using a substantial amount of human-annotated “instruction-response” data. Its core objective is to improve the model's ability to comprehend and adhere to natural language instructions, thereby strengthening its generalisation capacity and practical effectiveness in specific scenarios. In recommendation tasks, instruction tuning typically involves the following four steps:

- Step 1 (Define the Recommendation Task Instruction):** Clarify the objective of the recommendation task and convert it into a natural language instruction.
- Step 2 (Construct Input-Output Pairs):** Build the task inputs and outputs based on the user's historical behaviour.
- Step 3 (Form Instruction Tuning Samples):** Combine the task instruction with the task input to create an instruction input, and use the task output as the instruction output.
- Step 4 (Construct the Instruction Dataset):** Organise the formatted samples into a standard format (*Instruction Input*, *Instruction Output*), which is then used for instruction tuning of the model.

An example of formatting recommendation data for instruction tuning is shown below.

Instruction Input
Task Instruction: Given the user's historical interactions, please determine whether the user will like the target new movie by answering “Yes” or “No”.
Task Input: User liked the movie: The Godfather. User disliked the movie: Titanic. Target new movie: The Godfather Part II.
Instruction Output
Task Output: Yes.

4 The Proposed PRAG Method

This section provides a detailed description of our proposed PRAG Framework. First, we present an overview of the PRAG framework. Next, we introduce how to compute the semantic similarity between historical interaction items and target items. Finally, we explain how to perform fine-tuning using prompt-instruction pairs augmented with similarity information.

4.1 An Overview for the PRAG Method

An overview of the PRAG method is illustrated in Figure 3. The PRAG framework primarily consists of three main steps.

Step I (Computing Semantic Similarity): Brief descriptions of historical interaction items and target items (e.g., product descriptions in e-commerce data; if such descriptions are unavailable, they can be generated by an LLM) are encoded into embeddings using an embedding model. The similarity between these embeddings is then computed.

Step II (Constructing Dynamic Prompts): The semantic similarity scores calculated between each user's historical interaction items and the target item are dynamically incorporated into the prompt.

Step III (Fine-Tuning with Similarity Information): The LLM is fine-tuned using instruction pairs augmented with semantic similarity information, enabling the LLM-based recommender system to more effectively capture personalised user interest patterns.

4.2 Computing Semantic Similarity

In certain application scenarios, such as e-commerce platforms, product description texts are typically readily available, enabling direct semantic similarity computation based on such content. However, for datasets containing only item names without corresponding descriptions, large language models can be leveraged to automatically generate relevant item introductions. For instance, when dealing with a movie dataset lacking plot summaries, we can employ the following prompt template to generate movie summaries using an LLM:

• **Prompt template.** #

Task Instruction: You are a professional movie critic providing concise movie summaries. Provide a brief summary of the input movie focusing on its plot, characters, and themes. Keep it to about 100 words.

Task Input: The Godfather.

Given the generated (or original) summaries, we compute the semantic similarity between a target item i and a set of preferred items $\mathcal{P}$ as follows. First, we obtain the dense embedding vectors for the target summary S_i and each preferred summary S_j , $j \in \mathcal{P}$ using a pretrained sentence embedding model $\mathcal{E}(\cdot)$. That is,

$$H_i = \mathcal{E}(S_i), \quad H_j = \mathcal{E}(S_j), \quad j \in \mathcal{P}. \quad (1)$$

where $|\mathcal{P}|$ is the number of preferred items with valid summaries.

We then compute the cosine similarity between the target embedding and each preferred embedding:

$$\text{sim}(i, j) = \frac{H_i^T H_j}{\|H_i\| \|H_j\|}. \quad (2)$$

where $\text{sim}(i, j)$ denotes the semantic similarity scores.

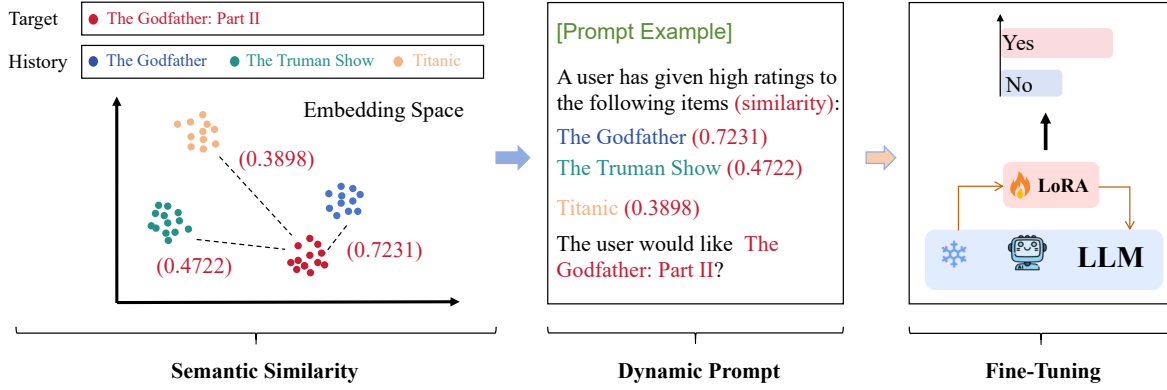


Figure 3: The architecture of our proposed PRAG method. PRAG first computes semantic similarity between users’ historical interaction items and target items. These similarity scores are then dynamically integrated as adaptive weights into prompt templates. Finally, instruction tuning is performed on the large language model using instruction sets augmented with semantic similarity information, thereby enhancing the model’s capability to capture users’ personalised preferences.

4.3 Constructing Dynamic Prompts

We adaptively integrate the calculated semantic similarity between historical interaction items and the target item as weights into the prompt. The final dynamic prompt format for each user is structured as follows:

• **Prompt template. #**

Task Instruction: A user has given high ratings to the following items (semantic similarity): The Godfather (0.7231), The Truman Show (0.4722), Titanic (0.3898). The user would enjoy the item titled The Godfather Part II. Answer with “Yes” or “No”.

Task Input: User’s liked items: The Godfather, The Truman Show, Titanic. User’s disliked items: Star Wars. Target new movie: The Godfather Part II

Task Output: Yes.

4.4 Fine-Tuning with Similarity Information

Large language models typically contain massive parameters, and performing full-parameter fine-tuning directly would require enormous computational resources and time costs, which are often unaffordable for most small and medium-sized research institutions. Based on the finding that the knowledge of language models is primarily distributed in low-dimensional subspaces [28], PRAG adopts the parameter-efficient LoRA fine-tuning method [28], which significantly enhances model performance by optimising only a small number of parameters [27, 38, 43].

We use instructional data containing semantic similarity information between users’ historical interaction items and target items for model fine-tuning. Specifically, PRAG freezes all original parameters of the pre-trained language model and trains only the inserted low-rank matrices, thereby achieving efficient injection of task-relevant knowledge. The original LoRA training objective for

the LLM in the PRAG framework can be expressed as:

$$\mathcal{L}_{LoRA} = \max_{\Theta} \sum_{(X,Y) \in \mathcal{Z}} \sum_{t=1}^{|Y|} \log P_{\Phi+\Theta}(Y_t|X, Y_{<t}) \quad (3)$$

where Φ denotes the original parameters of the LLM $\mathcal{M}$, which remain frozen during training, while Θ represents the LoRA parameters, which are updated during the training process (constituting only about 0.1% of the original LLM parameters [28]). X denotes the instruction input, and Y denotes the instruction output. Y_t refers to the t -th token of Y , and $Y_{<t}$ refers to the tokens preceding Y_t .

4.5 Semantic-Driven Personalised Interest Modelling

The core idea of PRAG is to shift from traditional ID-driven modelling of personalised interest information to semantic-driven modelling of personalised interest information. To clearly illustrate this paradigm shift, we compare the key differences between the two approaches in the Table 2:

Specifically, given a user’s historical preferred item set $\mathcal{P}$ and a target item i , traditional methods attempt to model a function $f(\text{ID}_{\mathcal{P}}, \text{ID}_i)$. In contrast, PRAG achieves semantic-driven modelling through the following steps:

(1) **Semantic Representation:** We map each item from its ID to its rich textual description space, obtaining its semantic embedding H (as shown in Eq. (1)).

(2) **Semantic Relation Quantification:** We compute the semantic similarity $\text{sim}(i, j)$ between the target item and each historical item (as shown in Eq. (2)). This score quantifies their relevance at the content level.

(3) **Personalised Prompt Construction:** We incorporate these similarity scores as explicit weights into the prompts, directly guiding

Table 2: Comparison of Traditional ID-Driven Paradigm vs. PRAG Semantic-Driven Paradigm

Dimension	Traditional ID-Driven Paradigm	PRAG Semantic-Driven Paradigm
Item Representation	Discrete, meaningless IDs	Continuous, semantically rich text embedding vectors
Relation modelling	Statistical associations: based on user-item interactions	Semantic associations: based on item description content
Personalisation Basis	Group behaviour patterns: (“users who liked A also liked B”)	Individual semantic reasoning: (“because the user liked A, and A is similar to B in content”)
Cold-start Capability	Weak, relies on interaction data	Strong, can recommend directly based on content similarity
Interpretability	Weak, typically “black-box” recommendations	Strong, similarity scores directly reveal recommendation rationale

the LLM to focus on the user’s history that is most semantically relevant to the target item, thereby achieving deep and interpretable personalised reasoning.

Theoretical Underpinnings of Semantic Similarity Modelling. The core rationale for employing semantic similarity, as opposed to traditional ID-based or collaborative signals, is rooted in its intrinsic alignment with the fundamental operating principles of LLMs, namely attention mechanisms and representation learning. The self-attention and cross-attention mechanisms in Transformers, which form the computational backbone of LLMs, inherently compute a form of similarity (e.g., dot-product) between query and key vectors to determine contextual relevance. Our method of computing semantic similarity between a target item and a user’s historical items, and injecting these scores as explicit weights into the prompt, can be viewed as a coarse-grained, pre-computed attention bias. It externally guides the LLM’s internal attention to focus more on the historically interacted items that are semantically most relevant to the current target, thereby streamlining the model’s reasoning process and enhancing its focus. In contrast, discrete IDs lack a native semantic space, forcing the LLM to learn their meanings from scratch during fine-tuning, which is inefficient and struggles with generalisation.

5 Experiments

In this section, we perform extensive experiments to address the following research questions:

RQ1: Compared to existing LLM-based recommender systems, does our proposed PRAG framework demonstrate superior performance under both few-shot and full-shot conditions?

RQ2: How does the performance of our proposed PRAG framework vary when integrated into LLM-based recommender systems built upon different LLM backbones?

RQ3: How is the performance of PRAG affected when personalised interest information is removed, i.e., when similarity information is excluded from the instructions?

5.1 Experimental Settings

We begin by introducing the datasets, followed by descriptions of the baseline methods and evaluation metrics. Lastly, we present the parameter settings used for the PRAG model.

Table 3: Statistics of the evaluation datasets.

Dataset	#Train	#Valid	#Test
movie	8,000	1,000	1,000
book	19,416	2,427	2,427
ML-1M	33,891	10,401	7,331
Amazon-Book	727,468	25,747	25,747

Dataset Description. We conduct experiments on four real-world datasets: movie [17], book [76], ML-1M [23] and Amazon-Book [24], summarised in Table 3. Details of the datasets can be found in Appendix A.

Few-shot Training Setting. In real-world applications, data sparsity in specific domains makes few-shot training research critically important. Under this setting, models are fine-tuned using only a minimal number of training samples (i.e., “shot”, e.g., 16) to simulate extreme data-constrained conditions. This aims to investigate whether PRAG can effectively unleash the recommendation potential of LLMs even with limited training data. To validate the effectiveness of our proposed framework under few-shot conditions, we conducted systematic experiments on both movie and book datasets, evaluating whether PRAG can still enhance the performance of LLM-based recommender systems under data scarcity.

Baselines. To verify the effectiveness of our PRAG framework, we integrate it into two existing LLM-based recommender systems and compare it with the following baseline methods:

(1) **GRU4Rec** [26]: GRU4Rec employs recurrent neural networks (RNN) to model complete user session sequences, aiming to capture dynamic preferences in sequential recommendation tasks, thereby achieving more accurate prediction. (2) **Caser** [58]: Caser utilises convolutional neural networks (CNN) to perform convolutional operations on users’ recent interaction sequences, effectively capturing their short-term interests and local sequential patterns, thereby achieving unified modelling of both long-term and short-term user preferences. (3) **SASRec** [34]: SASRec leverages self-attention mechanisms to model user behaviour sequences, adaptively assigning weights to historical interactions in order to dynamically capture information most relevant to the current recommendation. By integrating short-term interactions with long-term semantic dependencies, this mechanism effectively enhances the accuracy of sequential

Table 4: The AUC performance comparison of different methods under the few-shot scenario is shown below. Bold and underline are used to highlight the best and second-best results, respectively. “Rel. Imp.” indicates the average performance improvement rate of PRAG-TALLRec over the baseline model across three different shot settings.

Dataset Few-shot Methods	movie				book			
	16	64	256	Rel. Imp.	16	64	256	Rel. Imp.
GRU4Rec	0.4907	0.4987	0.5289	39.9%	0.4895	0.4964	0.4986	26.1%
Caser	0.4968	0.5106	0.5420	37.1%	0.4984	0.4972	0.4957	25.6%
SASRec	0.5043	0.5048	0.5225	38.7%	0.4948	0.5006	0.5020	25.0%
DROS	0.5076	0.5154	0.5407	35.9%	0.4928	0.4913	0.4913	26.9%
GRU-BERT	0.5085	0.5165	0.5344	36.3%	0.5007	0.4964	0.4979	25.2%
DROS-BERT	0.5021	0.5171	0.5394	36.2%	0.5007	0.4898	0.5020	25.5%
TALLRec	<u>0.6724</u>	<u>0.6748</u>	<u>0.7198</u>	2.8%	<u>0.5636</u>	<u>0.6039</u>	<u>0.6438</u>	3.4%
PRAG-TALLRec	0.6925	0.6961	0.7357	-	0.5914	0.6237	0.6573	-

recommendations. (4) **DROS** [67]: DROS integrates a distributional adaptation mechanism within the Empirical Risk Minimisation (ERM) framework. By explicitly simulating potential distributional discrepancies between training and testing environments, this approach enables models to proactively address distribution shifts, thereby significantly enhancing their generalisation capability and robustness on unseen data. (5) **GRU-BERT**: GRU-BERT enhances the GRU4Rec architecture by integrating the pretrained language model BERT [15], enabling synergistic modelling of item textual information and sequential structures. By deeply fusing semantic embeddings from BERT with item ID embeddings, the model constructs more comprehensive item representations, ultimately improving recommendation quality through more accurate capture of sequential dynamics. (6) **DROS-BERT**: DROS-BERT deeply integrates BERT’s advanced text modelling capabilities into the distributional adaptation framework of DROS. By leveraging textual semantic information to enhance the modelling and adaptation capacity for distribution shifts, the model synergistically improves both robustness and generalisation performance in scenarios with mismatched training-test data distributions. (7) **MF** [37]: MF decomposes the rating matrix into two low-dimensional latent factor matrices representing users and items, and utilises their inner product to model latent user-item interactions for preference prediction. (8) **LightGCN** [25]: LightGCN is a lightweight graph neural network recommendation model that employs simplified graph convolution operations to capture high-order interactions between users and items, thereby optimising representation learning. (9) **ICL** [12]: ICL leverages natural language prompts to activate the in-context learning capabilities of LLMs, aligning their recommendation outputs with instructional intent. This approach enables recommendation strategies to dynamically adapt to prompts enriched with background knowledge, user preferences, and behavioural instructions. (10) **Prompt4NR** [74]: Prompt4NR constructs diverse prompt templates (including fixed templates and learnable soft prompts) and enhances recommendation robustness and accuracy by aggregating predictions from these multiple templates to form consensus. Prompt4NR-V implements this approach using Vicuna-7B as its backbone LLM. (11) **TALLRec** [4]: TALLRec reframes recommendation tasks as instructional inputs for large language

models and optimises LLM-based recommender systems through instruction tuning, demonstrating exceptional performance particularly in few-shot scenarios. Building upon this foundation, we incorporate the PRAG framework to enhance its capability of modelling personalised interest information, naming this integrated variant PRAG-TALLRec. Following TALLRec’s few-shot training configuration, PRAG-TALLRec also employs LLaMA-7B as its backbone LLM. (12) **CoLLM** [73]: CoLLM enhances recommendation performance by injecting collaborative signals into LLM inputs, with its variant CoLLM-MF employing MF as the core collaborative module. Building upon this framework, we integrate the PRAG architecture into CoLLM-MF to strengthen its personalised interest modelling capability. This enhanced model is named PRAG-CoLMF and follows CoLLM’s configuration by adopting Vicuna-7B as its backbone LLM.

Evaluation Metrics. We evaluate the performance of recommendation methods based on two metrics: AUC and UAUC [50]. To ensure model robustness, each method is independently tested with five different random seeds, and the mean values of the performance metrics are reported.

Implementation and Configuration. We implement PRAG using PyTorch and optimise the LLM with the AdamW optimiser [36]. Details of the datasets can be found in Appendix B.

5.2 Performance Comparison

5.2.1 Few-shot Training Setting (RQ1). To validate the effectiveness of the proposed PRAG framework in enhancing recommendation performance of LLM-based recommender systems under data-constrained scenarios, we evaluate its performance under the few-shot setting. Table 4 presents the comparative results between our method and baseline models, leading to the following key observations:

(1) Under the few-shot setting, PRAG-TALLRec outperforms all baseline models in terms of performance. This demonstrates that with the integration of the PRAG framework, LLM-based recommender systems can effectively enhance their ability to model personalised user interests even with limited training samples, leading to significant improvements in recommendation effectiveness.

Table 5: The performance comparison of different methods under the full-shot setting is presented below. The best and second-best results are highlighted in bold and underlined, respectively. “Rel. Imp.” reflects the average improvement of PRAG-CoLMF over the baseline across the two evaluation metrics.

Dataset	ML-1M			Amazon-Book		
Methods	AUC	UAUC	Rel. Imp.	AUC	UAUC	Rel. Imp.
MF	0.6482	0.6361	12.2%	0.7134	0.5565	14.6%
LightGCN	0.5959	0.6499	15.7%	0.7103	0.5639	14.2%
SASRec	0.7078	0.6884	3.2%	0.6887	0.5714	15.5%
ICL	0.5320	0.5268	36.1%	0.4820	0.4856	50.4%
Prompt4NR-V	0.7071	0.6739	4.4%	0.7224	0.5881	11.1%
CoLLM-MF	<u>0.7295</u>	<u>0.6875</u>	1.7%	<u>0.8109</u>	<u>0.6225</u>	1.6%
PRAG-CoLMF	0.7423	0.6992	-	0.8235	0.6322	-

(2) Under few-shot scenarios, traditional recommendation models like GRU4Rec often underperform due to insufficient generalisation capability caused by data sparsity. This limitation prevents them from achieving deep semantic understanding and demonstrating strong generalisation performance—capabilities that LLM-based recommender systems maintain even under data-constrained conditions.

(3) Although LLM-based recommender systems possess strong generalisation capabilities for few-shot learning, they struggle to effectively capture users’ personalised interests. This limitation ultimately constrains their performance, leading to only suboptimal results.

5.2.2 Full-shot Training Setting (RQ1). To validate the effectiveness of LLM-based recommender systems integrated with the PRAG framework under sufficient training data conditions (full-shot), we conducted evaluations of their performance on the ML-1M and Amazon-Book datasets. The relevant results are presented in Table 5, with the main findings as follows:

(1) PRAG-CoLMF consistently outperforms all baselines on both evaluation metrics, demonstrating that the integration of the PRAG framework enables large LLM-based recommender systems to more comprehensively model users’ personalised interest information, thereby effectively enhancing the recommendation performance.

(2) LLM-based recommender systems demonstrate overall superior performance compared to traditional recommendation models, indicating their stronger capability in modelling complex semantic relationships.

(3) Although CoLLM-MF incorporates collaborative information for recommendations and outperforms LLM-based recommender systems that rely solely on ID information, it still falls short of our method. This indicates that existing LLM-based recommender systems cannot adequately leverage users’ personalised interest information, thereby limiting further improvements in recommendation effectiveness.

5.2.3 Overall Performance Analysis (RQ2). We integrated the PRAG framework into recommender systems based on different LLM backbones (such as LLaMA-7B and Vicuna-7B). Experimental results demonstrate that both PRAG-TALLRec and PRAG-CoLMF outperform traditional LLM-based recommender systems. Furthermore, the PRAG framework consistently achieves superior performance

under both few-shot and full-shot training settings. This indicates that our proposed PRAG framework exhibits strong generalisation capability, enabling flexible integration into existing LLM-based recommender systems and effectively enhancing their ability to model users’ personalised interests.

Additionally, we conducted an ablation study (RQ3), with detailed analysis provided in Appendix C.

Discussion. To illustrate the applicability of PRAG beyond recommendation scenarios, we provide a concrete sentiment analysis example with a dynamic prompt template in Appendix D.

6 Conclusion

In this work, we propose PRAG, a Retrieval-Augmented Generation framework tailored for predictive tasks. PRAG addresses a core limitation of conventional RAG, its reliance on retrievable ground truth, by computing the semantic similarity between a user’s historical interactions and a target item, and injecting these similarities as adaptive prompt weights. This design explicitly encodes history–target semantic correlations and enhances an LLM’s capacity to model personalised preferences. PRAG integrates seamlessly with existing LLM applications; instantiated in recommender systems, it strengthens semantics-aware user interest modelling. Empirically, across two LLM backbones and four real-world datasets, PRAG delivers consistent and significant gains in recommendation accuracy.

Limitations and Future Work. While PRAG improves predictive reasoning in recommendation, it currently assumes reliable similarity estimation from text (or item metadata) and does not yet optimise for efficiency under very long histories. Future work will extend PRAG to broader predictive settings (e.g., sentiment or demand forecasting), incorporate multimodal signals (e.g., images, reviews, and structured attributes), study robustness to noisy or sparse histories, and explore lightweight retrieval–prompt pipelines for latency-sensitive applications. We also plan theoretical analysis of the weighting mechanism and online A/B testing to assess real-world impact.

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A Datasets Details.

The construction and usage details for each dataset are introduced as follows:

movie. To conduct experiments under the few-shot learning scenario, we follow the preprocessing pipeline adopted by TALLREC [4] and use the preprocessed MovieLens-100K dataset [17]. From the original dataset, we extract the most recent 10,000 user interactions and split them into training, validation, and test sets in an 8:1:1 ratio. For each user sample, we retain the 10 interactions preceding the target item as the user’s historical behaviour. In accordance with the TALLREC setup, interactions with ratings higher than 3 are treated as positive samples (i.e., “like” behaviours).

book. We follow the preprocessing protocol of the BookCrossing dataset used in TALLREC [76]. For sample construction, we randomly select one interacted item as the target item for each user, and then randomly sample 10 other items from the remaining ones to form the user’s historical behaviour sequence. Subsequently, the entire dataset is divided into training, validation, and test sets with a ratio of 8:1:1, consistent with TALLREC. The dataset’s rating scale ranges from 1 to 10, and interactions with ratings higher than 5 are defined as positive samples (i.e., “liked” behaviours).

ML-1M. MovieLens-1M is a benchmark dataset widely used in the field of recommender systems, containing user ratings on movies on a 1-5 scale. We follow the data processing settings from the CoLLM paper: interactions with ratings above 3 are treated as positive samples (i.e., “liked” behaviours), and the dataset is constructed chronologically. Specifically, we select the most recent 20 months of interaction records, dividing them into the first 10 months for training, the subsequent 5 months for validation, and the final 5 months for testing. The final dataset comprises a total of 51,623 interactions from 839 users and 3,256 movies, with the training, validation, and test sets containing 33,891, 10,401, and 7,331 samples, respectively.

Amazon-Book. Amazon-Book dataset [24], a widely used book subset of Amazon review data in recommendation system research, contains user ratings on books ranging from 1 to 5. We strictly adhere to CoLLM’s configuration for data processing: interactions with ratings no less than 4 are treated as positive samples (i.e.,

Table 6: The ablation study of our PRAG method on two real-world datasets. The best results are in boldface.

dataset	few-shot	method	AUC
movie	16	w/o PRAG	0.6701
		PRAG-TALLRec	0.6925
	64	w/o PRAG	0.6713
		PRAG-TALLRec	0.6961
	256	w/o PRAG	0.7142
		PRAG-TALLRec	0.7357
book	16	w/o PRAG	0.5598
		PRAG-TALLRec	0.5914
	64	w/o PRAG	0.5996
		PRAG-TALLRec	0.6237
	256	w/o PRAG	0.6401
		PRAG-TALLRec	0.6573

“liked” behaviours). For data partitioning, we select all interaction records from 2017, using the first 11 months as the training set while equally dividing the last month’s data into validation and test sets. The final dataset covers 22,967 users and 34,154 books, containing 778,962 interaction records in total (training set: 727,468; validation set: 25,747; test set: 25,747).

B Implementation and Configuration

We implement PRAG using PyTorch and optimise the LLM with the AdamW optimiser [36]. The learning rate is selected from [1e-2, 1e-3, 1e-4], and the weight decay is set to 1e-3. We use the Llama3-8B to generate summaries and use the bge-large-en-v1.5 [65] embedding model from BAAI to encode the textual summaries. For PRAG-TALLRec, we follow the design of TALLRec and use the LLaMA-7B. The LoRA module is configured with the following hyperparameters: $r = 8$, $\alpha = 16$, $\text{dropout} = 0.005$, and target_modules is set to “[q_proj, v_proj]”. For the PRAG-CoLMF model, we follow CoLLM’s design and adopt Vicuna-7B as the base large language model. In module configuration, the hidden layer dimension of the multilayer perceptron in the CIE module is set to 10 times the input dimension, while the LoRA module maintains the same parameter settings as PRAG-TALLRec to ensure methodological consistency.

C Ablation Study.

To validate the effectiveness of the PRAG framework, we conducted ablation experiments under the k -shot training setting. Specifically, PRAG-TALLRec was compared with its variant (w/o PRAG), where

the model degenerates into a traditional LLM-based recommender system by removing personalised interest information (semantic similarity) from the instructions. As shown in Table 6, the experimental results demonstrate that:

(1) The performance of the model degrades after removing personalised interest information (w/o PRAG) and consistently underperforms compared to PRAG-TALLRec. This demonstrates that when semantic similarity information between user historical interactions and target items is lacking, LLMs struggle to effectively capture users’ personalised interests during fine-tuning, consequently impairing recommendation effectiveness. These results further validate the effectiveness of the PRAG framework in enhancing the performance of LLM-based recommender systems.

(2) Across two datasets and three k -shot training settings, PRAG-TALLRec consistently outperforms the “w/o PRAG” variant. This result robustly demonstrates the strong generalisation capability of the PRAG framework in adapting to varying degrees of data scarcity encountered in real-world scenarios. Even with limited training samples, the framework effectively enhances the ability of LLM-based recommender systems to model users’ personalised interests.

D Discussion.

The proposed PRAG framework is theoretically applicable to various predictive tasks, such as recommender systems and sentiment analysis. However, current empirical studies focus solely on recommendation scenarios, which may limit the generalisability of the findings. Future work will validate the effectiveness of the PRAG framework across a wider range of predictive tasks to more comprehensively assess its practical potential and scope of application. For example, in sentiment analysis tasks, we can use PRAG to construct the following dynamic prompt:

• Prompt template.

Task Instruction: A user has previously expressed the following sentiments (semantic similarity to the target text): “I absolutely love this brand’s customer service” (0.7652), “The battery life is a bit short” (0.4531). Predict whether the user’s new review is positive: “The new update made the interface so much smoother.” Answer “Yes” or “No”.

Task Input: User’s historical comments: “I absolutely love this brand’s customer service”, “The battery life is a bit short”. Target new comment: “The new update made the interface so much smoother.”

Task Output: Yes.